A3-PS007-TC112

VEIL6

Privacy-Aware IPv6 IoT Security Monitoring Across Address Rotation

Team CIPHER NEST

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Problem Description

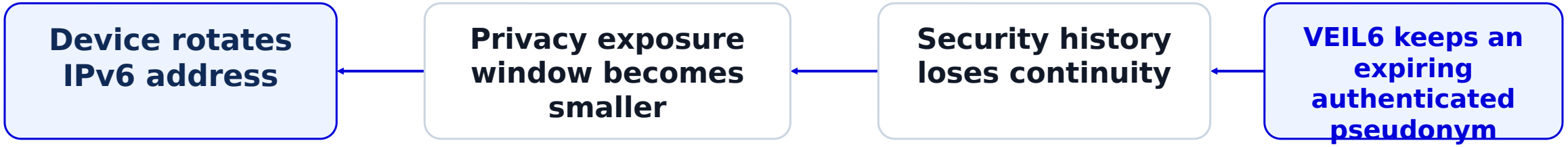
Temporary IPv6 addresses protect privacy, but they also break simple address-based security tracking.

- IoT devices rotate IPv6 addresses, so one device appears under several short-lived addresses.
- Monitoring tools may lose continuity or create a permanent identity map that weakens privacy.
- Rogue Router Advertisements and abnormal Neighbor Discovery traffic can still threaten the local link.
- The challenge is to preserve security context during an authorized session and erase the link when it expires.

Goal: measurable detection without permanent device tracking

Background

IPv6 privacy addresses reduce long-term correlation by changing interface identifiers. Local security still needs short-lived context to interpret Router Advertisement and Neighbor Discovery events.



Design boundary

A packet observation alone does not prove device identity. VEIL6 creates mappings only after authenticated enrollment, scopes them by tenant and removes them at expiry.

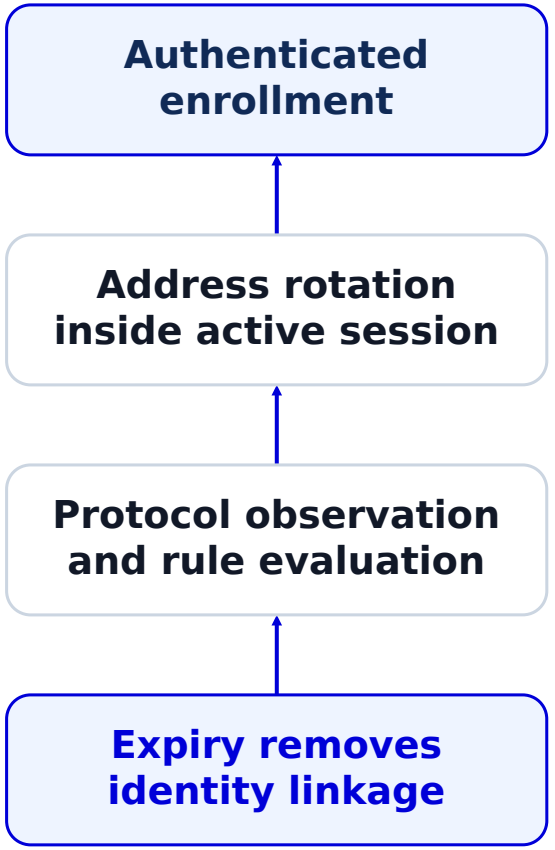
Literature & RFC Review

Reference	What it establishes	VEIL6 design decision
RFC 8981	Temporary IPv6 addresses change over time to reduce address-based correlation.	Use rotation as a privacy feature and avoid stable identifiers.
RFC 4861	Neighbor Discovery supports router discovery, address resolution and reachability on the local link.	Observe RA, NS and NA traffic with packet-level timestamps.
RFC 7113	Forged Router Advertisements remain a practical threat and RA-Guard can face evasion conditions.	Compare RA origin with an authorized-router policy and retain evidence.

Research gap The standards define privacy addresses and IPv6 control traffic, but not a complete expiring identity context for a small local IoT monitor.

Proposed Solution

- Create an isolated IPv6 lab with five emulated IoT clients and one authorized router.
- Authenticate each client locally and issue a random tenant-scoped pseudonym with a configurable expiry.
- Update the active mapping when the device rotates its temporary IPv6 address.
- Detect rogue Router Advertisements and abnormal Neighbor Discovery rates with saved packet evidence.
- Erase expired mappings and reject cross-tenant or replayed lookups.



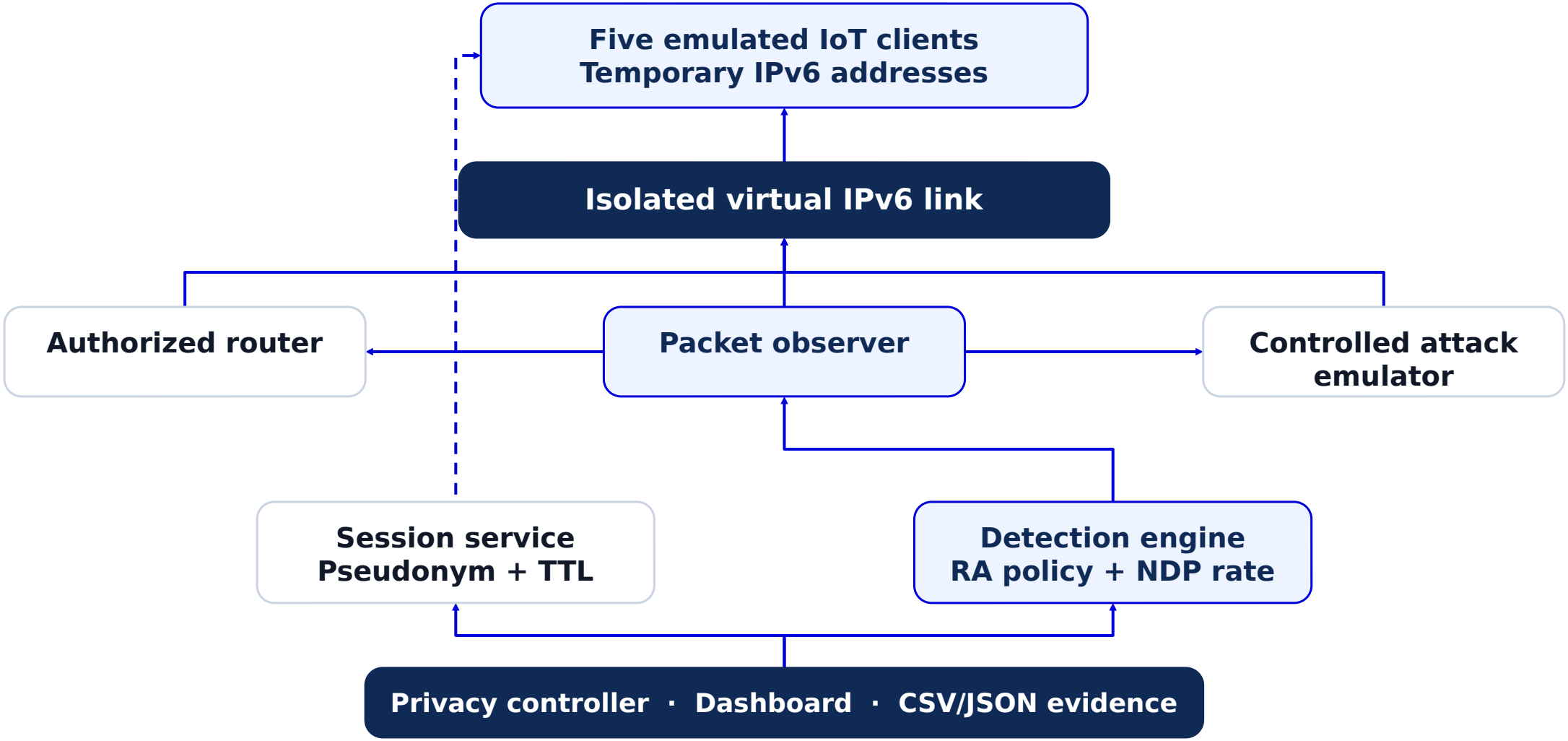
Optimization Proposed

Baseline approach	VEIL6 optimization	Expected improvement
Track only the IPv6 address	Bind changing addresses to a random pseudonym during an authenticated session	Continuity during rotation without a permanent identifier
Fixed NDP threshold	Calibrate a normal baseline and compare it with a static threshold	Measured false positives and transparent tuning
Keep mappings indefinitely	Use a configurable 10-minute demo TTL and verify lookup failure after expiry	A testable privacy boundary
Single shared namespace	Scope pseudonyms and lookups by tenant	No cross-tenant correlation in controlled tests

Evaluation: detection rate, false-positive rate, detection latency, rotation continuity and expiry compliance

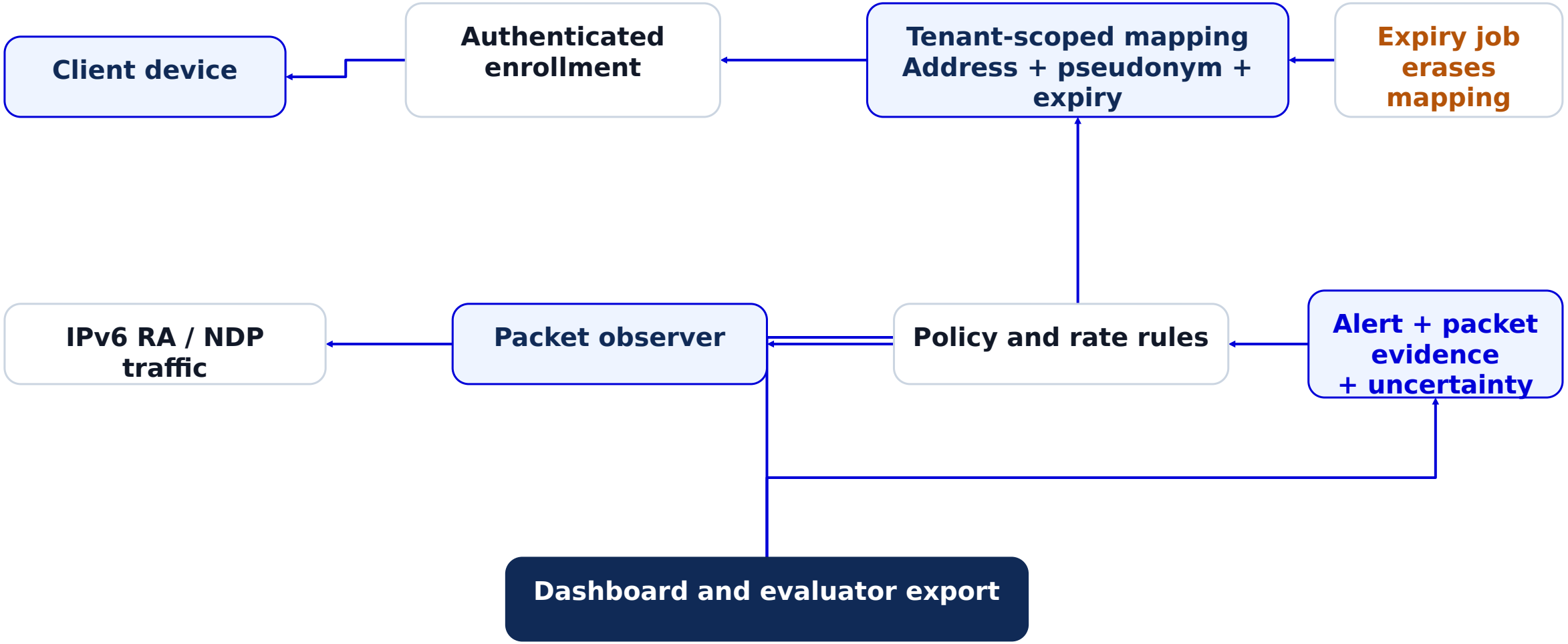
Solution Architecture & Design

Controlled IPv6 lab and local privacy-aware monitoring pipeline



Data Flow Diagram

Two evidence paths meet only inside an active authenticated session



Deployment Timeline

Window	Build milestone	Review gate
8-11 Oct	Repository, threat model, isolated IPv6 link, five clients and authenticated sessions	Real IPv6 traffic and repeatable address rotation
12-14 Oct	Level 1 presentation, team video and repository link	Steps 1-7 complete before organizer deadline
16-23 Oct	Packet observer, rogue RA rule, NDP baseline and metrics exporter	T03-T06 evidence with packet captures
24-28 Oct	Tenant isolation, replay tests and repeated experiment runs	False positives, latency and expiry review
29-31 Oct	Mentor fixes, final source, live demo link and technical report	Independent rerun from README

Official gate: Level 1 Steps 1-7 by 14 October. Mentoring and Steps 8-10 run from 16-31 October.

Bibliography

[1] AIORI-3. PS-007: Privacy-Aware IPv6 IoT Security Monitoring Across Address Rotation. 2026.

[2] Gont, F., Krishnan, S., Narten, T., and Draves, R. RFC 8981: Temporary Address Extensions for Stateless Address Autoconfiguration in IPv6. 2021.

[3] Narten, T., Nordmark, E., Simpson, W., and Soliman, H. RFC 4861: Neighbor Discovery for IP version 6. 2007.

[4] Gont, F. RFC 7113: Implementation Advice for IPv6 Router Advertisement Guard. 2014.

Implementation note

Thresholds, performance values and privacy claims will be reported only after controlled experiments with labelled ground truth.

Thank You

Questions and discussion

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